G. SAI GIREESH REDDY

AI/ML Engineer | Generative AI
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EDUCATION

Marwadi University, Rajkot, Gujarat **2024 – 2028**
B.Tech in Computer Science and Engineering (Artificial Intelligence & Machine Learning) *CGPA: 8.0/10*

TECHNICAL SKILLS

Languages: Python, C++,Java, SQL, JavaScript

ML & Deep Learning: PyTorch, scikit-learn, NumPy, Pandas, neural networks, RNNs/LSTMs, transformers, attention, model evaluation

Generative AI: RAG, embeddings, vector retrieval, hybrid search, reranking, LangGraph orchestration, structured output & schema validation, LLM evaluation

Backend & Data: FastAPI, PostgreSQL, pgvector, row-level security, Supabase, REST APIs, Next.js

Tools: Git, GitHub, Docker, Linux, CI/CD , n8n

PROJECTS

CodeAxis — Codebase Intelligence & Agent Context | *Tree-sitter, NetworkX, FastAPI, Next.js* GitHub

- Designing and building a platform that takes a repository link, archive or local folder and turns an unfamiliar codebase into something a developer or an AI agent can reason about, across three pillars: **Codebase Intelligence**, exploring structure and dependencies through a unified Component Inspector; an **Agent Context Engine** supplying coding agents the relevant code, architecture context and preserved design constraints for a task; and a **Virtual Development Environment** turning a described change into a system workflow, component mapping and proposed file structure.
- Built the analysis layer on deterministic static analysis rather than model inference — GitPython repository traversal, Tree-sitter AST parsing, and a NetworkX graph of cross-file symbol and dependency edges — so every explanation is grounded in real source spans and dependency paths, staying traceable as repositories outgrow the model's context window.

SecureRAG — Permission-Aware Multimodal Retrieval | *LangGraph, BGE-M3, pgvector, PaddleOCR*

- Building a retrieval-augmented QA system over mixed sources — PDFs parsed with PyMuPDF, scanned images recovered via PaddleOCR, relational rows — enforcing document- and row-level access control at the retrieval layer, so content outside a user's permission scope is never retrieved, passed to the model or cited.
- Architected the pipeline end to end: JWT auth and a LangGraph orchestration layer routing between intent classification, vector retrieval and text-to-SQL, with BGE-M3 multilingual embeddings over one Supabase-hosted PostgreSQL instance combining pgvector, full-text search and row-level security.
- Designed a citation-verification stage that checks each generated claim against its cited evidence and refuses to answer on failure, making refusal the default failure mode instead of an unsupported answer.

MedConnect — Automated Clinical Intake & Triage Routing | *Gemma 4, Next.js, Maps API, PostgreSQL*

- Built an end-to-end patient-intake pipeline converting a free-text symptom description into a validated structured clinical record and routing each case to the appropriate hospital, escalating emergencies to direct-contact alerts — under a hard safety constraint that the system never diagnoses, only structures and routes information for clinicians.
- Engineered the inference path for reliability rather than best-case output: schema validation on every model response, backed by a deterministic keyword classifier behind Gemma's emergency classification, so a malformed or low-confidence generation degrades safely instead of silently dropping an urgent case.
- Built an automated evaluation harness that runs in CI on every prompt and schema change, scoring each candidate against a frozen labelled set and blocking merges that regress emergency recall — making prompt changes a tested, versioned deployment step.

EXPERIENCE & ACHIEVEMENTS

- **AI Engineer & Team Lead, Team NxtGen.7** — OOSC 4.0, Opportunity Open Source Conference, IIIT Allahabad. Led the team as its AI engineer, owning the design and implementation of a Sovereign On-Premise Agentic AI Workbench.
- **Team Lead, Team Los Blancos** — AI Innovation Hackathon 2026: Build Real-World AI Solutions, Bharat Academix. Led the team to deliver GramMitra, a voice-driven platform making digital services usable for low-literacy rural users.
- **Team Lead** — Google "Build with Gemma 4", GDG Rajkot. Led the team to build MedConnect, a robust automated hospital patient-intake and triage routing system.
- **NextGen 2.0 Hackathon** — ACM Fremont / Soft Computing Research Society, 2026, for Lucent, a claim-verification engine returning cited, graded verdicts on digital content across social media.